Predictive evaluation with limited labels: shrinkage, same-item peer evidence, and an audited replication

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Abstract

We study how far a stateless, training-free predictor can go in the Predictive AI Evaluation Competition, and audit our own earlier claims. A symmetric Beta–Binomial shrinkage rule beats the official empirical-mean reference on all 13,200 eligible subject–benchmark pairs of 151 public benchmarks, eight acquisition splits and grouped nested validation: Brier ALC 0.236814 to 0.205984 (95% component interval for the difference $[-0.038426, -0.027214]$). Three richer isolated-evidence families do not improve on it reliably. We then exploit a property of the official interface that we verified in the released ingestion code: one global list of revealed labels reaches every prediction, and the acquisition/evaluation split is drawn per pair, so another subject may have revealed the very item being predicted. A pre-registered experiment (E1) adds that peer's residual to a benchmark-context prior. In platform-like cohorts of two subjects per benchmark, selected by leave-one-fold-out validation, it lowers Brier ALC by 0.004437 against the submitted rule (95% $[-0.005139, -0.003741]$, all eight seeds, 118 of 134 benchmarks) and by 0.009450 with four subjects. Without peer labels its output is exactly the original rule. On Codabench, the original rule (submission 1013718) scored a formative Brier ALC of 0.208365 and the E1 predictor (submission 1015136) scored *pending*; each formative score covers only nine sampled pairs and cannot separate effects of this size. We also reproduce the earlier submission's numbers to 10^{-16} and correct several audit findings. All candidates, failures and negative results are retained.

1 Task, estimand, and scope

The competition asks for the probability of an observed binary outcome for an AI subject and item, given visible attributes and a limited set of revealed labels [1]. Subjects are configurations, including the model, harness, and reasoning settings. Local benchmark names and internal subject IDs are used to construct evaluation partitions; the submitted predictor uses only the supplied dictionaries and anonymous benchmark identifiers.

At budgets $n \in \{0, 1, 3, 7, 15, 31\}$, let B_n be the mean squared error over all recorded evaluation responses of a subject–benchmark pair. Its score is

$$L = 0.1B_0 + 0.2B_1 + 0.2B_3 + 0.2B_7 + 0.2B_{15} + 0.1B_{31}. \tag{1}$$

The final estimand averages L equally across pairs, irrespective of their number of responses. Repeated observations are retained within each pair; they do not create additional equally weighted pairs. Lower scores are better. Acquisition and evaluation items are disjoint and budgets are nested.

Our results are local revalidation on a public collection. The public benchmarks had been inspected during an earlier sprint; the new partitions are not previously unobserved scientific data. A new

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evaluation program and a reviewer who did not inspect the new scores provide checks against shared implementation errors, not a guarantee against all forms of adaptive analysis. No local score in this paper is an official private-test score or ranking.

2 Independent reconstruction and provenance

The audit read the fifteen original analysis scripts, fourteen released result files, report, model, and retained development logs. Its principal evaluator imports none of those analysis modules and does not deserialize the original `pares.pkl`. It reads the original Parquet tables in bounded batches, aggregates counts, successes, and the first observation for each subject–item combination, and independently constructs eligibility and acquisition trajectories.

Frozen inputs. The local cache records the following dataset revision:

`54074453da7b9911be0f48e4c560be547bc84d21`

All 278 public response files match their Hugging Face SHA-256 download records; 501 supporting item, subject, and benchmark files were also hashed and match. The release manifest pins 890 necessary public files, including all public benchmark metadata, with a combined size of 5,354,371,887 bytes. The public baseline source matches upstream after newline normalization and has an identical Python AST. Data are obtained from the permitted measurement database [2]; no outside training data or paid inference service was used.

The old report cited an earlier revision. A historical manifest comparison found 32 public response paths absent from that older snapshot and one changed response file. This makes the old revision an insufficient reproduction instruction. It does not by itself invalidate the reported numbers: the 14,733 previously aggregated subject–benchmark rows match the independently recomputed successes and response counts exactly. Two additional all-zero pairs contain 116 responses in total and are ineligible because they have 60 and one distinct items. The old binary filter excluded them. The current binary scan contains 70,193,656 physical rows, of which 68,840,604 have an observed response; missing values must not be counted as labels.

Exact replication of the earlier design. Separately from the primary evaluation, we recreated the old six-pair and 150-evaluation-item sampling rules. Source row order was recovered directly from the response tables, rather than inferred from a hash-group output ordering. All 769 sampled pairs were reconstructed, including all 496 pairs in the original 101-benchmark headline set. Recomputed pair scores differ from the saved Beta–Binomial and empirical-mean scores by at most 1.11×10^{-16}. Their means are respectively 0.2104969827 and 0.2453845123. Thus the 14.2% headline gain is a reproducible result for that particular local design. It does not establish superiority to stronger competitors or to an unknown private-test distribution.

3 Audit findings and corrections

Better saved results were disclosed. The earlier report lists a subject-conditioned empirical-Bayes predictor at 0.20989535 and a macro hierarchical predictor at 0.21024007, both below the submitted predictor's point estimate. The former also heads the README table. Their uncertainty and runtime simplicity motivated the earlier choice. The retained artifacts provide no evidence of concealment. An audit cannot recover deleted or unsaved experiments or establish an author's intent.

Development isolation for richer priors was incomplete. The earlier LOBO procedure excluded the target benchmark but fitted a richer prior on every other benchmark, even during development selection. Its nominal headline set therefore contributed labels to those fitted priors. This is a legitimate transductive evaluation only if explicitly described as such; it is not a fully isolated development/headline partition. The constant Beta–Binomial rule fits no prior from those aggregates and does not directly inherit that fitted-prior defect. The new evaluator separates fitting, inner validation, and external testing by benchmark components.

Eligibility and weighting require distinct items. Ninety public pairs have at least 80 recorded responses but fewer than 80 distinct items. Some old descriptive statistics and empirical-prior filters

counted them as eligible. On the old headline benchmarks, response-based counting gives 11,657 pairs; the correct distinct-item criterion gives 11,639. The full new population contains 13,200 eligible pairs. The old cap of six subjects per benchmark changes the estimand and cannot be repaired exactly by merely reweighting those six observations. We remove both that cap and the 150-item evaluation cap.

Negative text experiments were narrower than their conclusions. The program called TF–IDF used word and character hashing without inverse document frequency. It predicted an item residual, evaluated against an in-sample empirical pair mean, with limited training rows and one optimizer configuration. Another text model used response-weighted samples and a budget-zero endpoint. These experiments are informative failures of those specific configurations; they do not establish that transferable item signal is absent, that pretrained representations cannot help, or that future improvement requires paid runtime language-model calls. We do not relabel them as independent successful replications in this audit.

Uncertainty and oracle language. The old interval $[-0.0373, -0.0249]$ belongs to a randomized acquisition run, not the headline trajectory. The original report distinguishes them. An independent bootstrap interval previously quoted for the headline artifacts has no reproducible generator in the released materials and is withdrawn. The original-protocol point estimates are retained as a replication; uncertainty for the new protocol is reported separately below. The older "sign-flip" routine was an exact binomial sign test, and zero opposite-sign bootstrap samples are not a true probability of zero. Finally, a retrospectively fitted per-item oracle is an empirical lower score on the observed sample; its within-item variance is not a demonstrated irreducible noise floor for new responses.

4 New evaluation and three initial improvements

For each eligible pair, items are canonically ordered and split 50/50 under eight SHA-256-derived seeds. One response is drawn uniformly from the recorded responses of each acquired item; equivalent Bernoulli draws from that item's empirical success fraction avoid expanding repeated records. All evaluation responses contribute to scoring. For any predictor constant within a pair, the exact sufficient-statistic loss is

$$B(p) = p^2 - 2pq + q, \qquad q = \frac{\text{evaluation successes}}{\text{evaluation responses}}. \tag{2}$$

This identity was checked against explicit squared errors on 150 synthetic samples. The primary trajectories are invariant to storage row order.

Exact shared content hashes connect the 151 evaluation benchmarks into 142 components. We keep each component within one of five external folds. Inside each external training set, another deterministic component split defines fitting and validation roles. Every feature transform and fitted model sees only the fitting role; every hyperparameter choice sees only inner validation. Public population imbalance is retained: for example, `alignment_faking` contributes 4,278 pairs and `sib200` contributes 3,341. Folds balance component counts, not pair counts.

The simple controls are the empirical mean and

$$\widehat{p} = \frac{k + 1.5}{n + 3}, \tag{3}$$

the posterior mean of a symmetric Beta prior. We claim no novelty for this standard shrinkage rule. The first three predeclared improvements are: (A) a Beta mean/strength grid; (B) a conditional table indexed by budget and success count, regularized and mixed with the original predictor; and (C) a ridge prior using visible subject attributes, with categorical and name token features. All validation candidates are retained. The table regularizer has weight sixteen simulated rows, approximately two training pairs across eight seeds, not sixteen independent subjects.

The original shrinkage rule beats the empirical mean by 0.030830 ALC, with a 95% component-bootstrap interval of $[-0.038426, -0.027214]$ and an improvement on 133 of 151 benchmarks. Table 1 shows why the small point improvements of A and B do not justify promoting either. C performs worse. Variation across acquisition seeds does not replace uncertainty about transfer to new benchmarks.

Table 1: Independent isolated-pair evaluation. All 13,200 eligible pairs and eight seeds; no cross-subject labels. Deltas compare with original Beta–Binomial. Intervals are component bootstrap intervals, averaging seeds before resampling.

Method	Brier ALC	Delta	95% interval for delta
Constant 0.5	0.250000		
Empirical-mean reference	0.236814		
Original Beta–Binomial	0.205984		
A: tuned Beta prior	0.205535	-0.000449	$[-0.005350, +0.003663]$
B: calibration table	0.205877	-0.000107	$[-0.000522, +0.000105]$
C: subject ridge prior	0.216549	$+0.010565$	$[+0.001218, +0.017943]$

Table 2: Conditional availability experiments. The same configuration selected in inner validation with cohorts of four is used in the other scenarios. These are separate simulated environments, not a choice of the best environment.

Subjects/cohort	Original API	Fallback extension	Delta	95% interval
1	0.205984	0.205984	0	exact probability equality
2	0.206010	0.204711	-0.001298	$[-0.003207, -0.000152]$
4	0.206027	0.203465	-0.002562	$[-0.005234, -0.000938]$
8	0.206030	0.202565	-0.003465	$[-0.006599, -0.001555]$

5 Conditional use of cross-subject evidence

After inspecting the initial experiments, we separately registered an exploratory extension motivated by the documented interface: supplied `labeled` entries can include different subject–benchmark pairs. The submitted program can therefore use another visible subject's acquired labels when their anonymous benchmark matches. It must never use an unseen evaluation label or assume a persistent shared cache.

For a target subject let (k, n) summarize its matching visible attributes, and let (k_o, n_o) summarize other supplied subjects on the same benchmark. Our fallback-preserving predictor uses

$$m_o = \frac{1.5 + k_o}{3 + n_o}, \quad m = 0.5 + \rho(m_o - 0.5), \quad \widehat{p} = \frac{k + 3m}{n + 3}. \tag{4}$$

With no peer evidence $m_o = m = 0.5$, so this is exactly the original predictor. A variant that also retunes strength is retained as a separate failed candidate: it loses this protective property and worsens isolated-pair loss.

We simulate deterministic random cohorts of four subjects per benchmark as the primary availability scenario, with sizes one, two, and eight as sensitivities. The final partial block can have fewer subjects. Both methods receive the same cohort's acquired labels at each budget. Eight visible attribute signatures are shared by 59 internal subject pairs; matching dictionary attributes, rather than inaccessible IDs, correctly pools those labels for both the original reference and the extension. Thus the cohort reference differs slightly from the isolated-ID simulation in Table 1; they must not be conflated.

For cohorts of four, the relative reduction is 1.24% against its correctly matched API reference, favorable in all eight acquisition seeds. Adjusting for the three initial candidates and the two contextual candidates gives a 99% interval of $[-0.006257, -0.000551]$. With retuned strength instead, ALC is 0.207251, a worse point estimate than the cohort reference. We retain this negative result. The exported fallback candidate uses $\rho = 0.5$, chosen by the pair-count-weighted mean of logged inner-validation scores at fixed strength three. The headline contextual result estimates the nested selection procedure; it is not a fresh unbiased test of a fixed parameter chosen after fitting to all public development data.

The simulator-to-function correspondence was checked independently on 5,376 synthetic cases across cohort sizes, mixtures, strengths, visible collisions, and partial blocks: maximum error zero. Another 528 valid (n, k) states confirm exact fallback. Probability-level equality is tested exactly; mathematically identical score reductions are canonicalized only after a 10^{-14} numerical check to avoid mistaking floating-point noise for a nonzero effect.

Table 3: E1 on held-out folds: Brier ALC difference, mean over benchmarks, 95% bootstrap over content components, seeds with a negative difference and benchmarks improved. CTX is the benchmark-context rule of Section 5.

Cohort	Comparison	Delta	95% interval	Seeds	Benchmarks
2	E1 vs submitted rule	-0.004437	$[-0.005139, -0.003741]$	8/8	118/134
2	E1 vs CTX	-0.001930	$[-0.002305, -0.001546]$	8/8	116/134
2	residuals without CTX vs CTX	$+0.000459$	$[-0.000390, +0.001357]$	0/8	72/134
4	E1 vs submitted rule	-0.009450	$[-0.010788, -0.008131]$	8/8	97/105
4	E1 vs CTX	-0.005583	$[-0.006602, -0.004598]$	8/8	92/105

6 What the platform actually supplies

When the experiments above were run, Codabench was unreachable and peer-label availability was an assumption. Two sources now settle it. First, the released streaming ingestion of the official baseline repository (`tools/streaming_ingestion.py`, commit `cc88a95`) receives one `labeled` list per checkpoint from the coordinator and passes that same list to every evaluation call of that budget; it is not filtered by subject or benchmark. Second, the rules state that the 50/50 acquisition/evaluation split is drawn *per subject–benchmark pair* [1]. Hence an item in one subject's evaluation pool can sit in another subject's acquisition pool and be revealed. Our first Codabench submission confirmed that samples contain benchmarks with more than one subject: its formative feedback lists nine pairs on seven benchmarks, two of which carry two subjects, each with 44–76 evaluated items (Table 4). We used this feedback only to size the simulation below, never to fit parameters, and we did not try to identify the anonymous subjects or benchmarks.

7 Same-item peer residuals (E1)

Hypothesis. Item difficulty is shared across subjects. A peer's label on the target item, measured against that peer's own rate, should move the target subject's prediction for that item in the same direction. The predictor of Section 5 cannot express this: it is constant within a pair, and the largest unexplained component of the error is variation between items inside a pair.

Predictor. Let m be the context prior of Section 5 and (k, n) the target subject's own labels on the benchmark. For each other subject j with (k_j, n_j) labels on the benchmark and a revealed label y_j on the target item, define the leave-one-out residual $r_j = y_j - (k_j - y_j + 1.5)/(n_j + 2)$. Then

$$\widehat{p} = \mathrm{clip}\Big(\frac{k + 3m}{n + 3} + \lambda \frac{\sum_j r_j}{c + \#\{j\}}\Big), \qquad \lambda = 0.75,\ c = 1. \tag{5}$$

Items are matched by their full supplied dictionary, subjects by their full visible attributes. With no same-item peer label the shift is zero; with no peer on the benchmark at all the output equals the original rule exactly.

Design. The hypothesis, grid ($\lambda \in \{0.25, 0.5, 0.75, 1\}$, $c \in \{0, 1, 3\}$, base rule with or without the context prior) and acceptance rule were written down before running. For every public benchmark with at least two eligible subjects, eight fixed seeds and up to three disjoint cohorts per benchmark, we draw two subjects (four as a sensitivity), restrict to the items both attempted, sample 88–152 of them as observed on Codabench, split each pair 50/50 independently, acquire the first 31 acquisition items at random and reveal one Bernoulli draw from each item's recorded success rate. At budget n every pair in the cohort has revealed n labels, and every prediction sees all of them. Scoring uses every recorded evaluation response, the official budget weights, pairs averaged within cohorts and benchmarks weighted equally. Hyperparameters are chosen on four of the five content-component folds and scored on the fifth. The configuration above was selected in all five folds for both cohort sizes. Acceptance required the 95% component-bootstrap interval of the held-out difference to lie below zero and the difference to be negative in at least six of eight seeds.

Result. E1 is accepted (Table 3, Figure 1). The gain over the context rule is larger with more subjects per benchmark, as expected if it comes from same-item evidence. Residuals alone, without

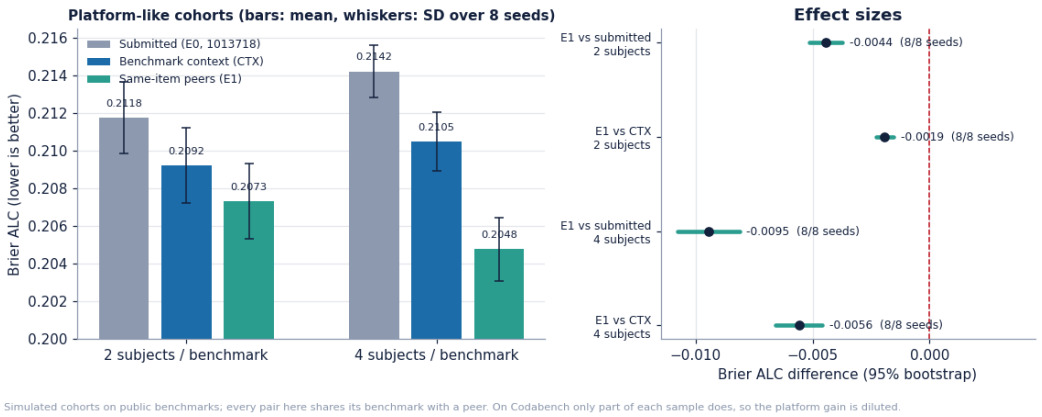


Figure 1: E1 in platform-like cohorts. Left: Brier ALC per configuration (mean and standard deviation over eight seeds). Right: held-out differences with 95% component-bootstrap intervals.

Table 4: Codabench formative feedback (phase "Competition", task "Amortized Prediction"). Each submission is scored on a newly sampled set of nine subject–benchmark pairs; across-pair standard error of the mean is about 0.022, so these numbers check execution, not the difference between methods.

ID	Archive	Pairs	Brier ALC	ECE
1013718	`submission_original.zip` (Beta–Binomial)	9	0.208365	0.136929
1015136	`submission_item_peer.zip` (E1)	–	*pending*	–

the context prior, are not better than the context rule with two subjects; both parts are needed. The shipped `model.py` reproduces the simulator's scores exactly (216 pair curves, maximum error zero) and returns exactly the original rule's probabilities when only the target subject's labels exist (2,220 checks). It passes the official `check_submission_zip.py`.

Expected size on the platform. Every simulated pair shares its benchmark with a peer. In the one Codabench sample we have seen, four of nine pairs did. If that fraction is typical, the platform gain is about $0.44 \times 0.0044 \approx 0.002$ Brier ALC against the submitted rule; this is an arithmetic projection, not a measurement.

8 Official Codabench results

Both archives ran to completion (Table 4). For submission 1013718, the formative Brier at budgets 0, 1, 3, 7, 15 and 31, averaged over its nine pairs, was 0.2500, 0.2284, 0.2142, 0.1975, 0.1869 and 0.1796. The per-pair tables are committed with the code. Our selected submission for final evaluation is 1015136, the method of Section 7; it falls back exactly to the rule of 1013718 whenever no peer evidence is supplied.

9 Inference, limitations, and reproducibility

We average the eight seeds within each pair, then bootstrap 10,000 complete content components. Each replicate recomputes total score difference divided by total pair count, preserving the official pair-weighted estimand rather than silently averaging benchmark means. Reported 95% intervals are descriptive; family-adjusted intervals are used when comparing promoted candidate families. The grouped sign-randomization calculation uses a plus-one correction and requires symmetry assumptions; it is not treated as an exact distribution-free statement about transfer.

This bootstrap conditions on fitted cross-validation models; it does not retrain the whole selection process for each replicate. Exact content hashes do not catch paraphrases or every family relationship. Public-population weights may differ from the platform's subject sampling, and binary outcomes

have heterogeneous meanings across released benchmarks. The inner validation sets and folds can be imbalanced by a few large benchmarks. Fitted models are not refitted on fit plus validation after selection, which is valid but may sacrifice efficiency. We make no claim to exhaust model classes or the potential value of pretrained representations.

The cohort simulator makes assumptions we could not check: the platform's subject sampling, the share of items that two sampled subjects have in common, and the use of one Bernoulli draw for an item with several recorded responses. Cohorts were built only from items attempted by every member, which favours same-item evidence; Section 7 gives the diluted projection. The private test benchmarks are disjoint from all public data, so the size of the transfer is unknown. E1 never uses an unrevealed label, a hidden identifier or cross-call state; it reads only the `labeled` argument of the current call.

How to rerun. The repository README lists every command. In short: `python audit/download_pinned.py -data DIR` fetches the 890 pinned files (5.35 GB) and checks their SHA-256; `audit/build_independent.py` builds the 13,200 pairs; `audit/evaluate_independent.py` and `audit/evaluate_context.py` produce Tables 1 and 2; `experiments/E1/run_e1.py` 2 and 4 produce Table 3 from the per-item cache in about one minute on a CPU, and `experiments/E1/parity_e1.py` checks the shipped predictor against the simulator. `build/` regenerates every figure and animation. Package versions are pinned in `requirements.txt` and `experiments/E1/requirements-e1.lock`. No GPU, network access at scoring time or paid inference is needed.

Code, protocols, results, the three submission archives and the formative feedback are public at `https://github.com/Agnuxo1/AIMS-PAIEC-2026` under the MIT licence; benchmark data keep their upstream licences and are not redistributed. The initial release is tagged `v1.0.0`; this report corresponds to release `v2.0.0`.

Training data and disclosure. The only data used for development are the public releases of the permitted measurement database at revision `54074453` [2]; no other source, no private test data and no paid inference were used, and the predictors fit nothing at scoring time. LLM-based coding assistants (Anthropic Claude, including Claude Code) were used for code, analysis scripts and writing, and as independent reviewers that did not see new scores before their review. All numbers were produced by deterministic scripts committed with the paper. Process failures are retained: one replication attempt assumed a hash-group output order, two checks needed JSON and floating-point fixes, and the first E1 analysis call failed on a column-name format and was re-run on unchanged simulation outputs. The earlier technical report withdrew a bootstrap interval without a reproducible generator. The same author previously developed the BenchClaw benchmarking tools; no code from them is used here.

References

[1] Stanford AIMS. Predictive AI Evaluation Competition at NeurIPS 2026. `https://aims.stanford.edu/competition/`. Accessed 28 October 2026.

[2] AIMS Foundations. Measurement database, frozen revision `54074453`. `https://huggingface.co/datasets/aims-foundations/measurement-db`.

[3] AIMS Foundations. PAIEC predictor baselines. `https://github.com/aims-foundations/paiec_baseline`.